C250 Aluminium Tooling Plate

Product Datasheet

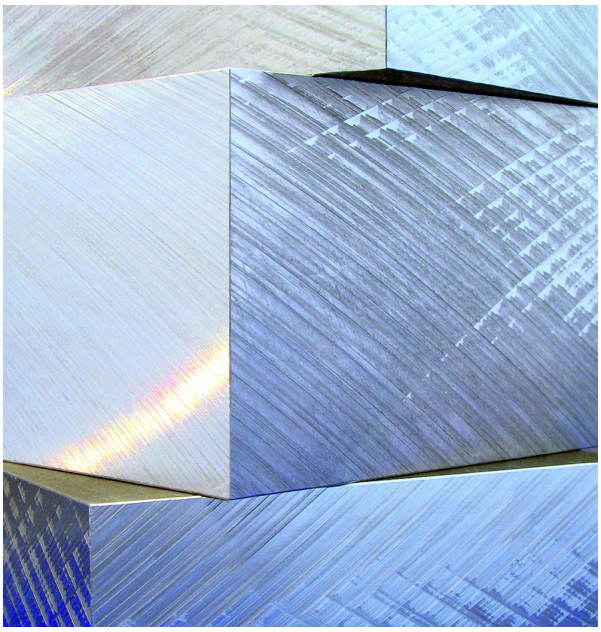
Cast Aluminium Tooling Plate

Service. Quality. Value.

Precision Milled Plate

Manufactured from 5083 aluminium alloy, C250 is a cast aluminium tooling plate product with a fine grain, homogenised structure.

The material offers excellent dimensional stability combined with high mechanical strength. C250 tooling plate is stress relieved while still able to retain 85% - 90% of its core strength when compared to rolled plate products. With excellent flatness, C250 provides enhanced corrosion resistance and retains excellent dimensional tolerances and flatness after machining. The plate product has excellent machining, welding and anodising capabilities, and both sides of the plate are foiled. This precision milled plate also benefits from extremely low residual stress. It is precision milled to a roughness Ra of 0.4 microns.



Key Features

- Excellent dimensional tolerances
- High strength
- Excellent flatness
- Superior corrosion resistance
- Extremely low residual stress
- Excellent weldability
- Excellent machining and anodising capabilities
- Homogenised and stress relieved
- Two surfaces, precision milled

Chemical Composition (weight %)

Weight (%)	Al	Mn	Fe	Cu	Mg	Si	Zn	Cr	Ti	Other (ea)	Other (total)
Min	Bal	0.40			4.00			0.05			
Max	Bal	1.00	0.40	0.10	4.90	0.40	0.10	0.25	0.25	0.05	0.15

Mechanical Properties (typical)

Tensile Strength	Proof Stress	Elongation A	Hardness (Brinell)
230 - 260 Mpa	110-130Mpa	10 - 15%	68 - 73 HB

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Quality & Testing:

ISO 9001 Quality Management Systems CERTIFIED

UKAS TESTING 1930

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